Internship

Report

Of

NetSage AI: Cisco Network Troubleshooting Assistant

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 2410031084

Name of Student: Abhay Shakya

Batch: 2024-28

CANDIDATE'S DECLARATION

I, **Abhay Shakya** do hereby declare that the internship report titled **"NetSage AI: Cisco Network Troubleshooting Assistant"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name :- **Abhay Shakya**
Roll No. :- **2410031084**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their inspiring guidance, constructive criticism, and valuable advice during this work. I am sincerely grateful to everyone who contributed to my learning and provided continuous encouragement throughout the internship.

I would like to thank the **AICTE Internship Program** for providing me with the opportunity and structured platform to gain practical exposure and develop industry-oriented skills. I am also thankful to my institute, **IILM University, Greater Noida**, and the faculty of the **School of Computer Science and Engineering** for their continuous support, guidance, and encouragement throughout the internship.

I express my gratitude to my parents for their blessings and constant support.

Signature

Student Name : **Abhay Shakya**

Roll No. : **2410031084**

ABSTRACT

NetSage AI: Cisco Troubleshooting Assistant is an AI-assisted network troubleshooting assistant designed to help users identify and resolve common Cisco networking problems using structured network evidence. The assistant combines local AI reasoning through Ollama with deterministic rule validation, human review, and post-fix verification to provide a controlled and explainable troubleshooting workflow.

The assistant accepts information such as network symptoms, topology details, show-command outputs, networking concepts, severity, and OSI-layer information. Using this evidence, the AI generates a probable root cause, confidence level, supporting evidence, next diagnostic command, and recommended corrective steps. The resulting diagnosis is then compared with an independent deterministic rule checker to identify agreement, conflict, or the absence of a deterministic finding.

A Human Review Gate allows the reviewer to accept, edit, or reject the recommendation before the case is considered complete. The assistant also includes a post-fix verification stage in which verification requires explicit evidence that the network problem has been resolved and connectivity has been restored. This prevents an AI recommendation from being treated as proof of a successful repair.

The project was implemented using Python, Ollama with the qwen3.5:4b local model, Streamlit, Cisco networking and Packet Tracer concepts, CSV-based case data, and automated testing with Pytest. The completed assistant contains a library of 44 troubleshooting cases and achieved 19 passing automated regression tests. The final implementation demonstrates a practical, human-in-the-loop approach to AI-assisted Cisco network troubleshooting.

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Internship Completion Certificate


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This certificate is awarded to

Abhay Shakya

for successfully completing

Introduction to Modern AI

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht

Swati Vashisht
Instructor
IILM University

13 Jun 2026
Completion Date

Cert ID: 44d412a4-8228-40fb-8cb9-dd4911896d90

Internship Offer Letter

Cisco Networking Academy <noreply@netacad.com> Sun, May 24, 2026 at 12:30 AM
To: abhay.shakya.cs28@iilm.edu

Cisco **Networking Academy**

Hello Abhay Shakya,

We have fantastic news! Swati Vashisht at IILM University has personally invited you to join

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Sincerely,

4. PROJECT DESCRIPTION

4.1 Introduction

NetSage AI is a local AI-assisted Cisco network troubleshooting application developed as part of the internship project. The system applies artificial intelligence to a practical networking problem: helping a user interpret network symptoms, identify likely causes, determine the relevant OSI layer and networking concept, and recommend evidence and Cisco diagnostic commands for further investigation.

The central design idea is to combine the flexibility of a language model with deterministic software safeguards. NetSage accepts a networking problem in structured case form or through the Streamlit interface, generates a structured diagnosis using a local AI model through Ollama, evaluates the recommendation through rule-based checks where structured configuration data is available, and then routes the result through a mandatory human-review gate. This makes the application a decision-support system rather than an autonomous network-management system.

A typical diagnosis contains the likely root cause, OSI layer, networking concept, severity, evidence to check, a recommended Cisco SHOW command, a proposed safe fix, confidence, and an explicit indication that human review is required. The final workflow also records review decisions and supports post-fix verification and reporting.

4.2 Organization Profile

The internship project was carried out in the context of the Cisco Networking Academy learning ecosystem and the student's academic program at IILM University. Cisco Networking Academy provides a practical environment for developing foundational and applied networking knowledge, while the associated "Introduction to Modern AI" learning context supported the project's focus on applying artificial intelligence to a real technical problem.

IILM University, Greater Noida, through the School of Computer Science and Engineering, provided the academic framework in which the internship project was planned, implemented and evaluated. The project duration covered May 2026 to August 2026. The internship work combined networking concepts, Python-based software development, AI integration, validation logic, user-interface development and project documentation.

4.3 Problem Statement

Troubleshooting a computer network requires more than identifying a possible fault. An engineer normally needs to interpret symptoms, map the issue to the appropriate networking layer, inspect configuration or command output, determine what evidence would confirm the hypothesis, and then choose a safe remediation step. For learners and junior engineers, this process can be difficult because the same symptom may have multiple possible causes across VLANs, trunking, routing, ACLs, NAT, interfaces, addressing and other network mechanisms.

The problem addressed by NetSage AI is therefore the lack of a structured, explainable troubleshooting assistant that can use AI to generate a diagnosis while still applying deterministic checks and keeping a human reviewer in control. The system should help organize the troubleshooting process, not blindly automate configuration changes.

4.4 Project Objectives

- Develop an AI-assisted troubleshooting workflow for common Cisco networking problems.
- Use Ollama with Llama 3.2 to generate structured network diagnoses from symptoms and evidence.
- Recommend relevant Cisco SHOW commands and evidence checks for diagnosis.
- Apply deterministic validation rules and compare AI recommendations with rule-based findings.
- Introduce a mandatory human-review gate with Accept, Edited/Needs Revision, and Reject decisions.
- Maintain review, verification and reporting records for transparent project evaluation.

4.5 Scope of the Project

The scope of NetSage AI covers AI-assisted diagnosis for common Cisco networking troubleshooting scenarios, especially issues that can be represented through symptoms, topology information, command evidence, and structured case data. The project supports the interpretation of faults across multiple OSI layers and networking concepts and provides a controlled workflow for reviewing recommendations before acceptance.

The implementation is suitable for academic demonstrations, Packet Tracer-based networking exercises, AI-assisted troubleshooting experiments, and evaluation of human-in-the-loop decision support. The system is not intended to directly modify production network devices or execute configuration changes automatically. Any proposed remediation remains subject to human authorization and verification.

4.6 Technologies and Tools Used

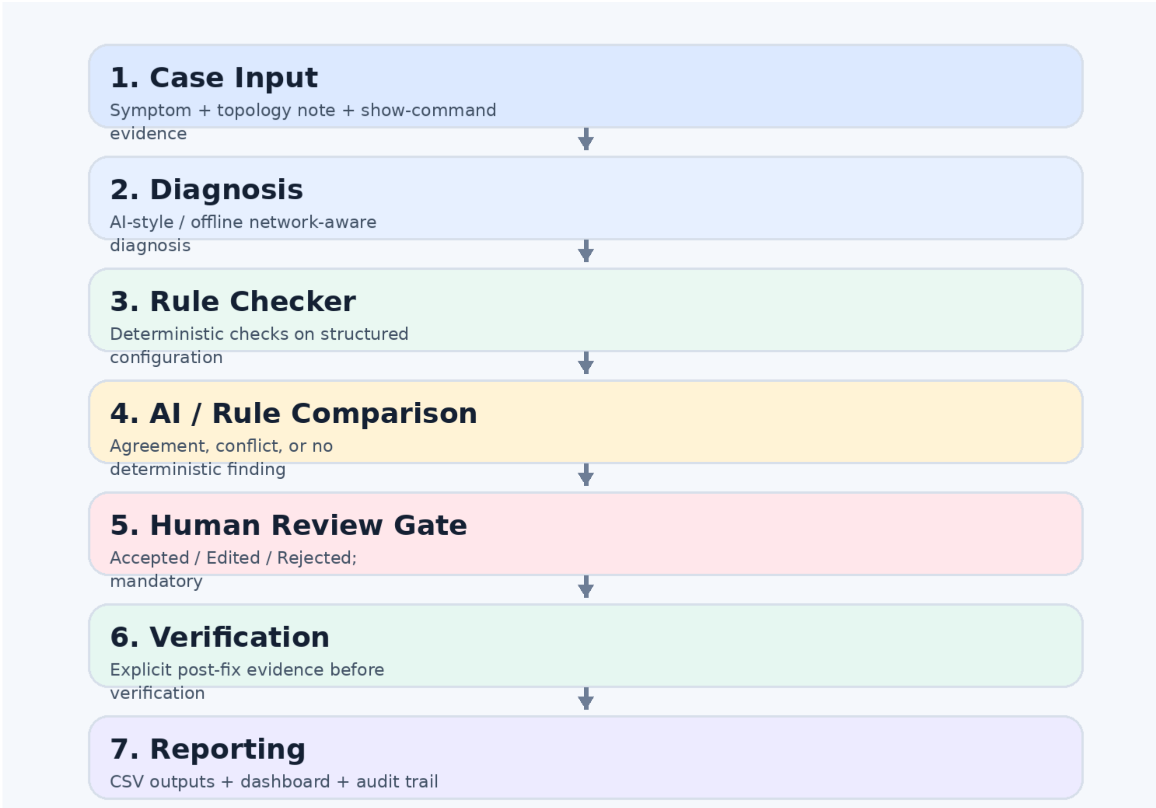
Technology / Tool	Purpose
Python	Core application logic, diagnosis pipeline, validation and reporting
Streamlit	Interactive web interface and review console
Ollama	Local serving layer for the AI model
Llama 3.2	Language model used for AI-assisted diagnosis
Cisco Packet Tracer	Network topology, lab scenarios and troubleshooting environment
Rule Checker	Deterministic checks for network configuration findings
JSON / Structured Schemas	Standardized diagnosis representation and validation
CSV / Logs	Cases, review decisions, results and audit information
Plotly / Dashboard Layer	Visual presentation of evaluation results

4.7 System Architecture

The NetSage architecture is organized as a sequence of decision-support stages. A user or test case provides the network problem, symptom, topology notes and available evidence. The AI diagnosis component then generates a structured recommendation. Where structured configuration data is present, the deterministic Rule Checker performs independent checks. The system compares the AI and deterministic outputs, routes the result to the human review gate, records the decision, and supports verification and reporting.

Figure 4.1: NetSage AI system architecture and troubleshooting workflow

The architecture deliberately separates diagnosis from evaluation. The expected fault in the reference cases is treated as ground truth for evaluation rather than being passed into the diagnosis engine to generate the answer. This separation reduces leakage between the model's diagnosis stage and the later evaluation stage.



4.8 Methodology

The project was developed through a phased methodology in which the diagnosis engine, deterministic checks, human-review workflow, verification and reporting layers were implemented and tested as separate components before being connected into the final pipeline. The methodology focused on keeping AI output structured, observable and reviewable.

Phase	Activity	Description
Phase 1	Problem definition	Identify common Cisco troubleshooting scenarios, required evidence, output fields and safety expectations.
Phase 2	Case and data preparation	Prepare structured troubleshooting cases and reference information for evaluation.
Phase 3	AI diagnosis	Generate independent structured diagnoses using the AI diagnosis engine and local Ollama model.
Phase 4	Deterministic validation	Apply rule-based checks to structured configuration information, including IP, subnet, gateway, interface, VLAN and route-related checks.
Phase 5	AI / rule comparison	Identify agreement, conflict or absence of deterministic findings and set the review status.
Phase 6	Human review	Allow the reviewer to Accept, Edit/Request Revision or Reject the recommendation. No network configuration is automatically applied.
Phase 7	Verification and reporting	Record verification outcomes and generate CSV outputs, audit logs, reports and a visual dashboard.

Key deterministic rules implemented in the Rule Checker include duplicate IP detection, wrong subnet mask detection, gateway mismatch checks, interface-down checks, missing VLAN checks and missing-route checks. These checks are intentionally deterministic and are used to complement, rather than replace, AI-generated diagnosis.

The human-review component records the reviewer decision and optional correction/comments. The review log therefore becomes part of the project's audit trail. The verification component can then evaluate post-fix evidence and record whether the expected network state has been restored.

Individual Contribution

My individual contribution focused on the data, validation and dashboard side of NetSage AI. This included work on the deterministic Rule Checker, test cases for validation rules, integration of validation results into the troubleshooting pipeline, and support for result reporting/dashboard outputs. The contribution was designed to strengthen the reliability of the AI workflow by providing checks outside the language model and by preserving reviewable evidence for evaluation.

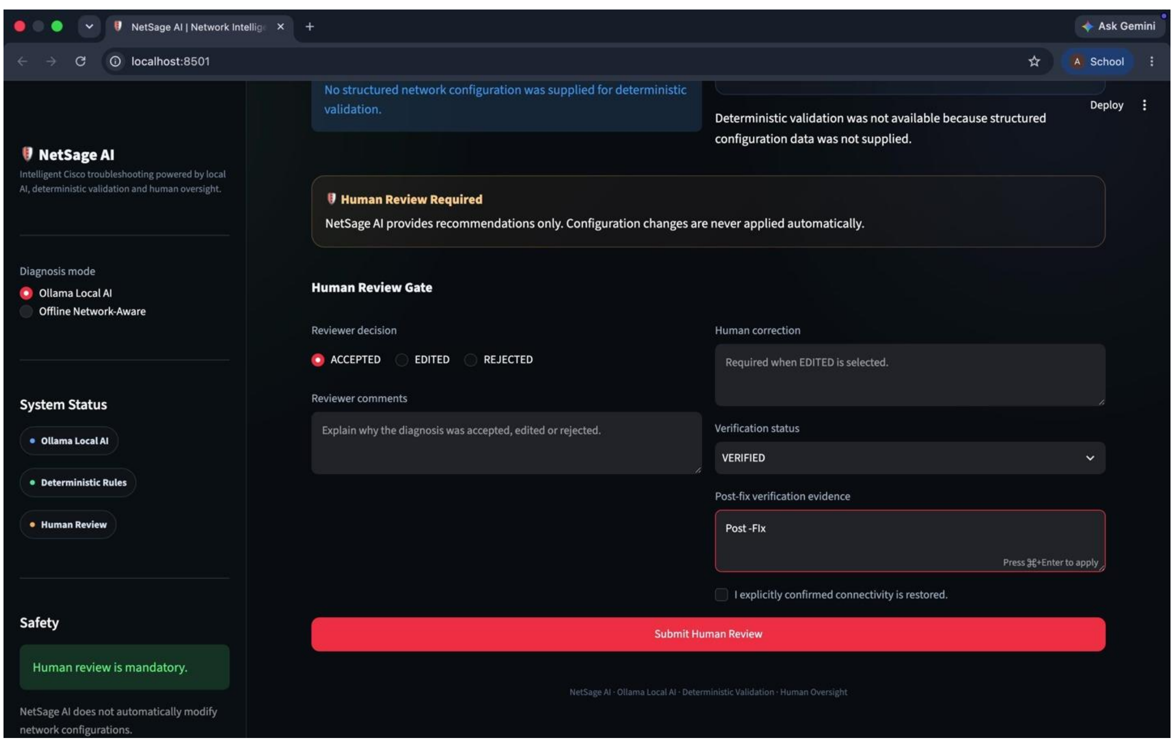
Project Evaluation and Test Evidence

The submitted project data contains 44 reference-based troubleshooting cases. In the available evaluation run, all 44 cases were recorded as accepted, with 34 medium-severity cases and 10 high-severity cases. The cases were distributed across Layer 1, Layer 2 and Layer 3 networking issues, with 23 Layer 2 cases, 20 Layer 3 cases and 1 Layer 1 case. The dashboard data therefore demonstrates broad coverage across common networking concepts.

The project also includes explicit human-review demonstration records for Accept, Edited and Rejected outcomes. These records are important because the design requirement is to keep a person in control of the final troubleshooting decision even when the AI recommendation is available.

Figure 4.2: Human Review Gate used to approve, revise or reject an AI diagnosis

Evaluation note: The 100% acceptance rate in the current reference-based evaluation run should not be reported as real-world AI accuracy. It reflects the available project test/evaluation records and should be interpreted within the limits of the reference dataset.



4.9 Expected Outcomes

- A working AI-assisted Cisco network troubleshooting application with a structured diagnosis output.
- Improved ability to relate symptoms to likely networking concepts and OSI layers.
- Recommended evidence checks and Cisco SHOW commands to support diagnosis.
- Deterministic validation outside the language model for structured network configuration checks.
- A mandatory human-review workflow that records acceptance, revision or rejection decisions.
- Post-fix verification and audit logging for traceability.
- Visual reporting of case volumes, severity, OSI-layer distribution and review outcomes.
- Practical understanding of combining AI with conventional software safeguards for responsible decision support.

Project Outputs

Output	Description
Application	Streamlit-based NetSage AI troubleshooting interface
AI Layer	Ollama + Llama 3.2 based structured diagnosis
Validation Layer	Deterministic rule checker and diagnosis validator
Human Review	Review gate with Accept / Edit / Reject decisions
Verification	Post-fix evidence verification module
Data Outputs	Case, review and evaluation CSV files
Reporting	Final report, dashboard and audit information
Network Lab	Cisco Packet Tracer topology file

4.10 Certificates of Completion and Communication Proof

As required by the IILM internship-report format, the original internship completion certificate and relevant official communication or offer-letter proof must be attached to the report. The current working copy contains explicit placeholder pages for these documents because the original certificate/communication files were not provided with the project source files used to prepare this report.

The project source repository and demo materials provide technical evidence of the NetSage implementation, including the application, AI diagnosis engine, rule-checking modules, test files, Packet Tracer topology, review logs, final dashboard data and demonstration recording.

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5. BIBLIOGRAPHY / REFERENCES

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2. Cisco Networking Academy, networking learning resources and Cisco Packet Tracer environment, accessed during the internship project.

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